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Namespace Uralstech.UDownloadManager

Classes

[AndroidDownload](#)

Represents a download managed by Android's `android.app.DownloadManager`.

[AndroidDownloadRequest](#)

Download request for Android.

[Download](#)

Represents a download.

[DownloadFilter](#)

(Android) Filters the downloads list.

[DownloadId](#)

Platform-agnostic download ID.

[DownloadManager](#)

[DownloadRequest](#)

Base class for download requests.

Enums

[DownloadFailReason](#)

Download failure reasons.

[DownloadPauseReason](#)

Download pause reasons.

[DownloadStatus](#)

Possible statuses for a download.

[NetworkTypes](#)

Allowed network types for a download.

[NotificationVisibility](#)

Download notification visibility settings.

Class AndroidDownload

Namespace: [Uralstech.UDownloadManager](#)

Represents a download managed by Android's `android.app.DownloadManager`.

```
public sealed record AndroidDownload : Download
```

Inheritance

object ← [Download](#) ← AndroidDownload

Inherited Members

[Download.Id](#) , [Download.Status](#)

Properties

Description

Gets the client-supplied description of the download, if any.

```
public string Description { get; }
```

Property Value

string

DownloadedSoFar

Gets the number of bytes downloaded so far.

```
public long DownloadedSoFar { get; }
```

Property Value

long

FailReason

Gets the reason the download failed.

```
public DownloadFailReason FailReason { get; }
```

Property Value

[DownloadFailReason](#)

Remarks

This value is only meaningful when [Status](#) is [Failed](#).

LastModified

Gets the UTC timestamp when the download was last modified.

```
public DateTime LastModified { get; }
```

Property Value

DateTime

MediaType

Gets the Internet media type (MIME type) of the downloaded file, if known.

```
public string? MediaType { get; }
```

Property Value

string

Path

Gets the local URI where the downloaded file is stored, or will be stored once the download begins.

```
public string? Path { get; }
```

Property Value

string

PauseReason

Gets the reason the download is paused.

```
public DownloadPauseReason PauseReason { get; }
```

Property Value

[DownloadPauseReason](#)

Remarks

This value is only meaningful when [Status](#) is [Paused](#).

Title

Gets the client-supplied title of the download, if any.

```
public string Title { get; }
```

Property Value

string

TotalSize

Gets the total size of the download, in bytes.

```
public long TotalSize { get; }
```

Property Value

long

Remarks

A value of **-1** indicates that the total size is not yet known.

Uri

Gets the URI from which the file is being downloaded.

```
public string Uri { get; }
```

Property Value

string

Methods

FromAndroidCursor(AndroidJavaObject)

Creates an array of [AndroidDownload](#) instances from an Android **Cursor** returned by `android.app.DownloadManager.Query`.

```
public static AndroidDownload[] FromAndroidCursor(AndroidJavaObject cursor)
```

Parameters

cursor AndroidJavaObject

The cursor containing one or more download records.

Returns

[AndroidDownload\[\]](#)

An array containing every download represented by the cursor. If the cursor contains no rows, an empty array is returned.

Exceptions

`InvalidOperationException`

Thrown if the cursor does not contain one of the required columns.

`PlatformNotSupportedException`

Thrown if this method is called on a runtime other than Android.

Class AndroidDownloadRequest

Namespace: [Uralstech.UDownloadManager](#)

Download request for Android.

```
public sealed class AndroidDownloadRequest : DownloadRequest
```

Inheritance

object $\leftarrow$ [DownloadRequest](#) $\leftarrow$ AndroidDownloadRequest

Constructors

AndroidDownloadRequest(string)

```
public AndroidDownloadRequest(string uri)
```

Parameters

uri string

The URI to download from.

Exceptions

PlatformNotSupportedException

Thrown if this constructor is called on a runtime other than Android.

Methods

AddRequestHeader(string, string)

Add an HTTP header to be included with the download request. The header will be added to the end of the list.


```
public void AddRequestHeader(string header, string value)
```

Parameters

header string

HTTP header name

value string

Header value.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

ObjectDisposedException

Dispose()

Disposes native Android instance.

```
public override void Dispose()
```

SetAllowedNetworkTypes(NetworkTypes)

Restrict the types of networks over which this download may proceed. By default, all network types are allowed. Consider using [SetAllowedOverMetered\(bool\)](#) instead, since it's more flexible.

```
public void SetAllowedNetworkTypes(NetworkTypes networkTypes)
```

Parameters

networkTypes [NetworkTypes](#)

Any combination of [NetworkTypes](#).

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

ObjectDisposedException

SetAllowedOverMetered(bool)

Set whether this download may proceed over a metered network connection. By default, metered networks are allowed.

```
public void SetAllowedOverMetered(bool allow)
```

Parameters

allow bool

Whether this download may proceed over a metered connection.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

ObjectDisposedException

SetAllowedOverRoaming(bool)

Set whether this download may proceed over a roaming connection. By default, roaming is allowed.

```
public void SetAllowedOverRoaming(bool allow)
```

Parameters

allow bool

Whether to allow a roaming connection to be used.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

ObjectDisposedException

SetDescription(string)

Set a description of this download, to be displayed in notifications (if enabled).

```
public void SetDescription(string description)
```

Parameters

description string

The description.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

ObjectDisposedException

SetDestination(string)

Set the destination for the downloaded file. Must be an absolute path on external storage, and the calling application may need the WRITE_EXTERNAL_STORAGE permission.

```
public void SetDestination(string path)
```

Parameters

path string

The path.

Remarks

By default, downloads are saved to a generated filename in the shared download cache and may be deleted by the system at any time to reclaim space.

For applications targeting Android 10 or above, WRITE_EXTERNAL_STORAGE permission is not needed and the **path** must refer to a path within the directories owned by the application (e.g. Application.persistentDataPath) or a path within the top-level Downloads directory (as returned by `Environment.getExternalStoragePublicDirectory(String)` with `Environment.DIRECTORY_DOWNLOADS`). All non-visible downloads that are not modified in the last 7 days will be deleted during idle runs.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

ObjectDisposedException

SetDestinationInExternalFilesDir(string, string?)

Set the local destination for the downloaded file to a path within the application's external files directory (as returned by `Context.getExternalFilesDir(String)`, aka Application.persistentDataPath).

```
public void SetDestinationInExternalFilesDir(string subPath, string? dirType = null)
```

Parameters

subPath string

The path within the external directory, including the destination filename.

dirType string

The directory type to pass to `Context.getExternalFilesDir(String)`

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

ObjectDisposedException

SetDestinationInExternalPublicDir(string, string)

Set the local destination for the downloaded file to a path within the public external storage directory (as returned by `Environment.getExternalStoragePublicDirectory(String)`).

```
public void SetDestinationInExternalPublicDir(string subPath, string dirType)
```

Parameters

subPath string

The path within the external directory, including the destination filename.

dirType string

The directory type to pass to `Environment.getExternalStoragePublicDirectory(String)`

Remarks

For applications targeting Android 10 or above, the `WRITE_EXTERNAL_STORAGE` permission is not needed and the **dirType** must be one of the known public directories like `Environment.DIRECTORY_DOWNLOADS`, `Environment.DIRECTORY_PICTURES`, `Environment.DIRECTORY_MOVIES`, etc.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

ObjectDisposedException

SetMimeType(string)

Set the MIME content type of this download. This will override the content type declared in the server's response.

```
public void SetMimeType(string mimeType)
```

Parameters

mimeType string

The MIME type.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

ObjectDisposedException

SetNotificationVisibility(NotificationVisibility)

Control whether a system notification is posted by the download manager while this download is running or when it is completed. If enabled, the download manager posts notifications about downloads through the system **NotificationManager**. By default, a notification is shown only when the download is in progress.

```
public void SetNotificationVisibility(NotificationVisibility visibility)
```

Parameters

visibility [NotificationVisibility](#)

The notification visibility.

Remarks

If set to [Hidden](#), this requires the permission **android.permission.DOWNLOAD_WITHOUT_NOTIFICATION**.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

ObjectDisposedException

SetRequiresCharging(bool)

Specify that to run this download, the device needs to be plugged in. This defaults to [false](#).

```
public void SetRequiresCharging(bool requiresCharging)
```

Parameters

requiresCharging bool

Whether the device is plugged in.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

ObjectDisposedException

SetRequiresDeviceIdle(bool)

Specify that to run, the download needs the device to be in idle mode. This defaults to [false](#).

```
public void SetRequiresDeviceIdle(bool requiresDeviceIdle)
```

Parameters

requiresDeviceIdle bool

Whether the device need be within an idle maintenance window.

Remarks

Idle mode is a loose definition provided by the system, which means that the device is not in use, and has not been in use for some time.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

ObjectDisposedException

SetTitle(string)

Set the title of this download, to be displayed in notifications (if enabled). If no title is given, a default one will be assigned based on the download filename, once the download starts.

```
public void SetTitle(string title)
```

Parameters

title string

The title.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

ObjectDisposedException

Class Download

Namespace: [Uralstech.UDownloadManager](#)

Represents a download.

```
public abstract record Download
```

Inheritance

object ← Download

Derived

[AndroidDownload](#)

Properties

Id

Gets the platform-agnostic identifier of the download.

```
public DownloadId Id { get; protected init; }
```

Property Value

[DownloadId](#)

Status

Gets the current status of the download.

```
public DownloadStatus Status { get; protected init; }
```

Property Value

[DownloadStatus](#)

Enum DownloadFailReason

Namespace: [Uralstech.UDownloadManager](#)

Download failure reasons.

```
public enum DownloadFailReason
```

Fields

CannotResume = 1008

Some possibly transient error occurred but we can't resume the download.

DeviceNotFound = 1007

No external storage device was found. Typically, this is because the SD card is not mounted.

FileAlreadyExists = 1009

The requested destination file already exists (the download manager will not overwrite an existing file).

FileError = 1001

A storage issue arose which doesn't fit under any other error code.

HttpDataError = 1004

An error receiving or processing data occurred at the HTTP level.

InsufficientSpace = 1006

There was insufficient storage space. Typically, this is because the SD card is full.

TooManyRedirects = 1005

There were too many redirects.

UnhandledHttpCode = 1002

An HTTP code was received that download manager can't handle.

Unknown = 1000

The download has completed with an error that doesn't fit under any other error code.

Class DownloadFilter

Namespace: [Uralstech.UDownloadManager](#)

(Android) Filters the downloads list.

```
public sealed class DownloadFilter
```

Inheritance

object $\leftarrow$ DownloadFilter

Methods

ByIds(params DownloadId[])

Filters by the given download IDs.

```
public void ByIds(params DownloadId[] ids)
```

Parameters

ids [DownloadId\[\]](#)

ByStatus(DownloadStatus)

Filters by the given download status.

```
public void ByStatus(DownloadStatus status)
```

Parameters

status [DownloadStatus](#)

Class DownloadId

Namespace: [Uralstech.UDownloadManager](#)

Platform-agnostic download ID.

```
public sealed record DownloadId
```

Inheritance

object ← DownloadId

Methods

FromAndroidId(long)

Creates a [DownloadId](#) from an Android-native ID.

```
public static DownloadId FromAndroidId(long id)
```

Parameters

id long

Returns

[DownloadId](#)

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

ToAndroidId()

Gets the Android-native ID.

```
public long ToAndroidId()
```

Returns

long

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

Class DownloadManager

Namespace: [Uralstech.UDownloadManager](#)

```
public sealed class DownloadManager : Singleton<DownloadManager>
```

Inheritance

object ← DownloadManager

Methods

Awake()

```
protected override void Awake()
```

CancelDownloads(params DownloadId[])

Cancel downloads and remove them from the download manager. Each download will be stopped if it was running, and it will no longer be accessible through the download manager. If there is a downloaded file, partial or complete, it is deleted.

```
public int CancelDownloads(params DownloadId[] ids)
```

Parameters

ids [DownloadId\[\]](#)

The IDs of the downloads to remove.

Returns

int

The number of downloads actually removed.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on an unsupported platform.

GetMobileDownloadSizeLimit()

Returns the maximum size of a download performed with a mobile connection, or -1 if there is no defined limit.

```
public long GetMobileDownloadSizeLimit()
```

Returns

long

Exceptions

PlatformNotSupportedException

Thrown if this method is called on an unsupported platform.

GetRecommendedMobileDownloadSizeLimit()

Returns the recommended maximum size of a download performed with a mobile connection, or -1 if there is no defined limit.

```
public long GetRecommendedMobileDownloadSizeLimit()
```

Returns

long

Exceptions

PlatformNotSupportedException

Thrown if this method is called on an unsupported platform.

QueryDownloads(DownloadFilter?)

Query the download manager about downloads that have been requested.

```
public IReadOnlyList<Download> QueryDownloads(DownloadFilter? filter = null)
```

Parameters

filter [DownloadFilter](#)

An optional filter.

Returns

IReadOnlyList<[Download](#)>

The downloads.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on an unsupported platform.

TryEnqueueDownload(DownloadRequest, out DownloadId?)

Tries to enqueue a new download. The download will start automatically once the download manager is ready to execute it and connectivity is available.

```
public bool TryEnqueueDownload(DownloadRequest request, out DownloadId? id)
```

Parameters

request [DownloadRequest](#)

The request. Must be of type [AndroidDownloadRequest](#) on Android.

id [DownloadId](#)

An ID for the download, unique across the system. This ID is used to make future calls related to this download. Returns -1 if the operation fails.

Returns

bool

[true](#) if successful, [false](#) otherwise.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on an unsupported platform.

Events

OnDownloadCompleted

Invoked when a download has completed, with its ID.

```
public event Action<DownloadId>? OnDownloadCompleted
```

Event Type

Action<[DownloadId](#)>

OnDownloadNotificationClicked

Invoked when the user has clicked on a download notification, with the ID of the download(s) the notification was for.

```
public event Action<DownloadId[]>? OnDownloadNotificationClicked
```

Event Type

Action<[DownloadId](#)[]>

OnViewDownloads

Invoked when the user has requested to see all downloads, with a boolean indicating if they should be sorted by size.

```
public event Action<bool>? OnViewDownloads
```

Event Type

Action<bool>

Enum DownloadPauseReason

Namespace: [Uralstech.UDownloadManager](#)

Download pause reasons.

```
public enum DownloadPauseReason
```

Fields

QueuedForWiFi = 3

The download exceeds a size limit for downloads over the mobile network and the download manager is waiting for a Wi-Fi connection to proceed.

Unknown = 4

The download is paused for some other reason.

WaitingForNetwork = 2

The download is waiting for network connectivity to proceed.

WaitingToRetry = 1

The download is paused because some network error occurred and the download manager is waiting before retrying the request.

Class DownloadRequest

Namespace: [Uralstech.UDownloadManager](#)

Base class for download requests.

```
public abstract class DownloadRequest
```

Inheritance

object ← DownloadRequest

Derived

[AndroidDownloadRequest](#)

Methods

Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public virtual void Dispose()
```

Enum DownloadStatus

Namespace: [Uralstech.UDownloadManager](#)

Possible statuses for a download.

```
[Flags]  
public enum DownloadStatus
```

Extension Methods

[AndroidInterop.ToAndroidDownloadStatus\(DownloadStatus\)](#)

Fields

Failed = 16

The download has failed (and will not be retried).

Paused = 4

The download is waiting to retry or resume.

Pending = 1

The download is waiting to start.

Running = 2

The download is currently running.

Successful = 8

The download has successfully completed.

Enum NetworkTypes

Namespace: [Uralstech.UDownloadManager](#)

Allowed network types for a download.

```
[Flags]  
public enum NetworkTypes
```

Fields

Mobile = 1

A Mobile data connection.

WiFi = 2

A Wi-Fi data connection.

Enum NotificationVisibility

Namespace: [Uralstech.UDownloadManager](#)

Download notification visibility settings.

```
public enum NotificationVisibility
```

Fields

Hidden = 2

This download doesn't show in the UI or in the notifications.

Visible = 0

This download is visible but only shows in the notifications while it's in progress.

VisibleNotifyCompleted = 1

This download is visible and shows in the notifications while in progress and after completion.

Namespace Uralstech.UDownloadManager. Native

Classes

[AndroidInterop](#)

Interface for the native Android plugin.

[AndroidInterop.DownloadCursor](#)

Partial interface for [Cursor](#)↗.

[AndroidInterop.DownloadCursor.Columns](#)

Possible columns for the cursor.

[AndroidInterop.DownloadManagerInterface](#)

The main native interface.

[AndroidInterop.DownloadManagerInterface.Callbacks](#)

Handles callbacks from the native interface.

[AndroidInterop.DownloadManagerRequest](#)

Interface for [DownloadManager.Request](#)↗.

[AndroidInterop.Uri](#)

Partial interface for [Uri](#)↗.

Enums

[AndroidInterop.DownloadCursor.FieldType](#)

Data type of column field.

[AndroidInterop.DownloadStatus](#)

Possible statuses for a download.

Class AndroidInterop

Namespace: [Uralstech.UDownloadManager.Native](#)

Interface for the native Android plugin.

```
public static class AndroidInterop
```

Inheritance

object ← AndroidInterop

Methods

ToAndroidDownloadStatus(DownloadStatus)

Converts a global download status to an Android download status.

```
public static AndroidInterop.DownloadStatus ToAndroidDownloadStatus(this  
DownloadStatus status)
```

Parameters

status [DownloadStatus](#)

Returns

[AndroidInterop.DownloadStatus](#)

ToDownloadStatus(DownloadStatus)

Converts an Android download status to a global download status.

```
public static DownloadStatus ToDownloadStatus(this AndroidInterop.DownloadStatus  
status)
```

Parameters

status [AndroidInterop.DownloadStatus](#)

Returns

[DownloadStatus](#)

Class AndroidInterop.DownloadCursor

Namespace: [Uralstech.UDownloadManager.Native](#)

Partial interface for [Cursor](#)🔗.

```
public static class AndroidInterop.DownloadCursor
```

Inheritance

object ← AndroidInterop.DownloadCursor

Methods

Close(AndroidJavaObject)

Closes the Cursor, releasing all of its resources and making it completely invalid.

```
public static void Close(AndroidJavaObject native)
```

Parameters

native AndroidJavaObject

The native object.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

GetBlob(AndroidJavaObject, int)

Returns the value of the requested column as a byte array.

```
public static sbyte[] GetBlob(AndroidJavaObject native, int columnIndex)
```

Parameters

native `AndroidJavaObject`

The native object.

columnIndex `int`

The zero-based index of the target column.

Returns

`sbyte[]`

Exceptions

`PlatformNotSupportedException`

Thrown if this method is called on a runtime other than Android.

GetColumnIndex(AndroidJavaObject, string)

Returns the zero-based index for the given column name, or -1 if the column doesn't exist.

```
public static int GetColumnIndex(AndroidJavaObject native, string columnName)
```

Parameters

native `AndroidJavaObject`

The native object.

columnName `string`

The column name.

Returns

`int`

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

GetDouble(AndroidJavaObject, int)

Returns the value of the requested column as a double.

```
public static double GetDouble(AndroidJavaObject native, int columnIndex)
```

Parameters

native AndroidJavaObject

The native object.

columnIndex int

The zero-based index of the target column.

Returns

double

Remarks

The result and whether this method throws an exception when the column value is [null](#), the column type is not a floating-point type, or the floating-point value is not representable as a double value is implementation-defined.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

GetFloat(AndroidJavaObject, int)

Returns the value of the requested column as a float.

```
public static float GetFloat(AndroidJavaObject native, int columnIndex)
```

Parameters

native AndroidJavaObject

The native object.

columnIndex int

The zero-based index of the target column.

Returns

float

Remarks

The result and whether this method throws an exception when the column value is [null](#), the column type is not a floating-point type, or the floating-point value is not representable as a float value is implementation-defined.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

GetInt(AndroidJavaObject, int)

Returns the value of the requested column as an int.

```
public static int GetInt(AndroidJavaObject native, int columnIndex)
```

Parameters

native AndroidJavaObject

The native object.

columnIndex int

The zero-based index of the target column.

Returns

int

Remarks

The result and whether this method throws an exception when the column value is [null](#), the column type is not an integral type, or the integer value is outside the range [`Integer.MIN_VALUE`, `Integer.MAX_VALUE`] is implementation-defined.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

GetLong(AndroidJavaObject, int)

Returns the value of the requested column as a long.

```
public static long GetLong(AndroidJavaObject native, int columnIndex)
```

Parameters

native AndroidJavaObject

The native object.

columnIndex int

The zero-based index of the target column.

Returns

long

Remarks

The result and whether this method throws an exception when the column value is [null](#), the column type is not an integral type, or the integer value is outside the range

[[Long.MIN_VALUE](#), [Long.MAX_VALUE](#)] is implementation-defined.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

GetShort(AndroidJavaObject, int)

Returns the value of the requested column as a short.

```
public static short GetShort(AndroidJavaObject native, int columnIndex)
```

Parameters

[native](#) AndroidJavaObject

The native object.

[columnIndex](#) int

The zero-based index of the target column.

Returns

short

Remarks

The result and whether this method throws an exception when the column value is [null](#)[↗], the column type is not an integral type, or the integer value is outside the range [[Short.MIN_VALUE](#), [Short.MAX_VALUE](#)] is implementation-defined.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

GetString(AndroidJavaObject, int)

Returns the value of the requested column as a string.

```
public static string GetString(AndroidJavaObject native, int columnIndex)
```

Parameters

native AndroidJavaObject

The native object.

columnIndex int

The zero-based index of the target column.

Returns

string

Remarks

The result and whether this method throws an exception when the column value is [null](#) or the column type is not a string type is implementation-defined.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

GetType(AndroidJavaObject, int)

Returns data type of the given column's value. The preferred type of the column is returned but the data may be converted to other types as documented in the get-type methods such as [GetInt\(AndroidJavaObject, int\)](#), [GetFloat\(AndroidJavaObject, int\)](#) etc.

```
public static AndroidInterop.DownloadCursor.FieldType GetType(AndroidJavaObject native, int columnIndex)
```

Parameters

native `AndroidJavaObject`

The native object.

columnIndex `int`

The zero-based index of the target column.

Returns

[AndroidInterop.DownloadCursor.FieldType](#)

Exceptions

`PlatformNotSupportedException`

Thrown if this method is called on a runtime other than Android.

IsClosed(AndroidJavaObject)

Returns [true](#) if the cursor is closed.

```
public static bool IsClosed(AndroidJavaObject native)
```

Parameters

native `AndroidJavaObject`

The native object.

Returns

`bool`

Exceptions

`PlatformNotSupportedException`

Thrown if this method is called on a runtime other than Android.

IsNull(AndroidJavaObject, int)

Returns [true](#) if the value in the indicated column is [null](#).

```
public static bool IsNull(AndroidJavaObject native, int columnIndex)
```

Parameters

native AndroidJavaObject

The native object.

columnIndex int

The zero-based index of the target column.

Returns

bool

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

MoveToFirst(AndroidJavaObject)

Move the cursor to the first row.

```
public static bool MoveToFirst(AndroidJavaObject native)
```

Parameters

native AndroidJavaObject

The native object.

Returns

bool

Remarks

This method will return [false](#) if the cursor is empty.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

MoveToNext(AndroidJavaObject)

Move the cursor to the next row.

```
public static bool MoveToNext(AndroidJavaObject native)
```

Parameters

native AndroidJavaObject

The native object.

Returns

bool

Remarks

This method will return [false](#) if the cursor is already past the last entry in the result set.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

Class

AndroidInterop.DownloadCursor.Columns

Namespace: [Uralstech.UDownloadManager.Native](#)

Possible columns for the cursor.

```
public static class AndroidInterop.DownloadCursor.Columns
```

Inheritance

object ← AndroidInterop.DownloadCursor.Columns

Fields

BytesDownloadedSoFar

Number of bytes download so far.

```
public const string BytesDownloadedSoFar = "bytes_so_far"
```

Field Value

string

Description

The client-supplied description of this download. This will be displayed in system notifications. Defaults to the empty string.

```
public const string Description = "description"
```

Field Value

string

Id

An identifier for a particular download, unique across the system. Clients use this ID to make subsequent calls related to the download.

```
public const string Id = "_id"
```

Field Value

string

LastModifiedTimestamp

Timestamp when the download was last modified, in `System.currentTimeMillis()` (wall clock time in UTC).

```
public const string LastModifiedTimestamp = "last_modified_timestamp"
```

Field Value

string

LocalUri

Uri where downloaded file will be stored. If a destination is supplied by client, that URI will be used here. Otherwise, the value will initially be null and will be filled in with a generated URI once the download has started.

```
public const string LocalUri = "local_uri"
```

Field Value

string

MediaType

Internet Media Type of the downloaded file. If no value is provided upon creation, this will initially be null and will be filled in based on the server's response once the download has started.

```
public const string MediaType = "media_type"
```

Field Value

string

Reason

Provides more detail on the status of the download. Its meaning depends on the value of [Status](#). When [Status](#) is [Failed](#), this indicates the type of error that occurred. If an HTTP error occurred, this will hold the HTTP status code as defined in RFC 2616. Otherwise, it will hold one of [DownloadFailReason](#). When [Status](#) is [Paused](#), this indicates why the download is paused. It will hold one of [DownloadPauseReason](#). If [Status](#) is neither [Failed](#) nor [Paused](#), this column's value is undefined.

```
public const string Reason = "reason"
```

Field Value

string

Status

Current status of the download, as one of [AndroidInterop.DownloadStatus](#).

```
public const string Status = "status"
```

Field Value

string

Title

The client-supplied title for this download. This will be displayed in system notifications. Defaults to the empty string.

```
public const string Title = "title"
```

Field Value

string

TotalSizeBytes

Total size of the download in bytes. This will initially be -1 and will be filled in once the download starts.

```
public const string TotalSizeBytes = "total_size"
```

Field Value

string

Uri

URI to be downloaded.

```
public const string Uri = "uri"
```

Field Value

string

Enum

AndroidInterop.DownloadCursor.FieldType

Namespace: [Uralstech.UDownloadManager.Native](#)

Data type of column field.

```
public enum AndroidInterop.DownloadCursor.FieldType
```

Fields

Blob = 4

Float = 2

Integer = 1

Null = 0

String = 3

Class

AndroidInterop.DownloadManagerInterface

Namespace: [Uralstech.UDownloadManager.Native](#)

The main native interface.

```
public static class AndroidInterop.DownloadManagerInterface
```

Inheritance

object ← AndroidInterop.DownloadManagerInterface

Methods

CreateInstance(Callbacks)

Gets an instance of the native interface.

```
public static AndroidJavaObject  
CreateInstance(AndroidInterop.DownloadManagerInterface.Callbacks callbacks)
```

Parameters

callbacks [AndroidInterop.DownloadManagerInterface.Callbacks](#)

Returns

AndroidJavaObject

A new instance of the native interface.

Remarks

This method should generally be called only once during the application's lifetime, as the native object is supposed to act like a singleton. If you create a new instance of the native object while another one is active, all unfulfilled callbacks meant for the existing instance will be forwarded to the new instance's **callbacks** instead.

The returned `AndroidJavaObject` should be disposed when no longer needed.

Exceptions

`PlatformNotSupportedException`

Thrown if this method is called on a runtime other than Android.

`Enqueue(AndroidJavaObject, AndroidJavaObject)`

Enqueue a new download. The download will start automatically once the download manager is ready to execute it and connectivity is available.

```
public static long Enqueue(AndroidJavaObject native, AndroidJavaObject request)
```

Parameters

native `AndroidJavaObject`

The native plugin instance.

request `AndroidJavaObject`

A `DownloadManager.Request` object.

Returns

`long`

An ID for the download, unique across the system. This ID is used to make future calls related to this download. Returns -1 if the operation fails.

Exceptions

`PlatformNotSupportedException`

Thrown if this method is called on a runtime other than Android.

`GetMaxBytesOverMobile(AndroidJavaObject)`

Returns maximum size, in bytes, of downloads that may go over a mobile connection; or [null](#) if there's no limit.

```
public static long? GetMaxBytesOverMobile(AndroidJavaObject native)
```

Parameters

native AndroidJavaObject

The native plugin instance.

Returns

long?

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

GetRecommendedMaxBytesOverMobile(AndroidJavaObject)

Returns recommended maximum size, in bytes, of downloads that may go over a mobile connection; or [null](#) if there's no recommended limit. The user will have the option to bypass this limit.

```
public static long? GetRecommendedMaxBytesOverMobile(AndroidJavaObject native)
```

Parameters

native AndroidJavaObject

The native plugin instance.

Returns

long?

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

Query(AndroidJavaObject, long[]?, DownloadStatus)

Query the download manager about downloads that have been requested.

```
public static AndroidJavaObject Query(AndroidJavaObject native, long[]? ids = null,
    AndroidInterop.DownloadStatus status = DownloadStatus.None)
```

Parameters

native AndroidJavaObject

The native plugin instance.

ids long[]

If not [null](#) or empty, include only the downloads with the given IDs.

status [AndroidInterop.DownloadStatus](#)

If not [None](#), include only downloads with status matching any of the given status flags.

Returns

AndroidJavaObject

A **Cursor** object.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

Remove(AndroidJavaObject, long[])

Cancel downloads and remove them from the download manager. Each download will be stopped if it was running, and it will no longer be accessible through the download manager. If there is a downloaded file, partial or complete, it is deleted.

```
public static int Remove(AndroidJavaObject native, long[] ids)
```

Parameters

native AndroidJavaObject

The native plugin instance.

ids long[]

The IDs of the downloads to remove.

Returns

int

The number of downloads actually removed.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

Class

AndroidInterop.DownloadManagerInterface.Callbacks

Namespace: [Uralstech.UDownloadManager.Native](#)

Handles callbacks from the native interface.

```
public sealed class AndroidInterop.DownloadManagerInterface.Callbacks
: AndroidJavaProxy
```

Inheritance

object ← AndroidInterop.DownloadManagerInterface.Callbacks

Constructors

Callbacks()

```
public Callbacks()
```

Exceptions

PlatformNotSupportedException

Thrown if this constructor is called on a runtime other than Android.

Events

OnDownloadCompleted

Broadcast intent action sent by the download manager when a download completes, along with the download ID.

```
public event Action<long>? OnDownloadCompleted
```


Event Type

Action<long>

OnDownloadNotificationClicked

Broadcast intent action sent by the download manager when the user clicks on a running download, either from a system notification or from the downloads UI, along with the download IDs.

```
public event Action<long[]>? OnDownloadNotificationClicked
```

Event Type

Action<long[]>

OnViewDownloads

Intent action to launch an activity to display all downloads, along with a boolean indicating if they should be sorted by size.

```
public event Action<bool>? OnViewDownloads
```

Event Type

Action<bool>

Class

AndroidInterop.DownloadManagerRequest

Namespace: [Uralstech.UDownloadManager.Native](#)

Interface for [DownloadManager.Request](#) .

```
public static class AndroidInterop.DownloadManagerRequest
```

Inheritance

object ← AndroidInterop.DownloadManagerRequest

Methods

AddRequestHeader(AndroidJavaObject, string, string)

Add an HTTP header to be included with the download request. The header will be added to the end of the list.

```
public static void AddRequestHeader(AndroidJavaObject native, string header, string value)
```

Parameters

native AndroidJavaObject

The native object.

header string

HTTP header name

value string

Header value.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

CreateInstance(AndroidJavaObject)

Creates a new Request object.

```
public static AndroidJavaObject CreateInstance(AndroidJavaObject uri)
```

Parameters

uri AndroidJavaObject

The HTTP or HTTPS URI to download.

Returns

AndroidJavaObject

A new instance of Request.

Remarks

The returned AndroidJavaObject should be disposed when no longer needed.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

SetAllowedNetworkTypes(AndroidJavaObject, NetworkTypes)

Restrict the types of networks over which this download may proceed. By default, all network types are allowed. Consider using [SetAllowedOverMetered\(AndroidJavaObject, bool\)](#) instead, since it's more flexible.

```
public static void SetAllowedNetworkTypes(AndroidJavaObject native,  
NetworkTypes networkTypes)
```

Parameters

native `AndroidJavaObject`

The native object.

networkTypes [NetworkTypes](#)

Any combination of [NetworkTypes](#).

Remarks

As of Android 7.0, setting only the [WiFi](#) flag here is equivalent to calling [SetAllowedOverMetered\(AndroidJavaObject, bool\)](#) with [false](#) [↗].

Exceptions

`PlatformNotSupportedException`

Thrown if this method is called on a runtime other than Android.

SetAllowedOverMetered(AndroidJavaObject, bool)

Set whether this download may proceed over a metered network connection. By default, metered networks are allowed.

```
public static void SetAllowedOverMetered(AndroidJavaObject native, bool allow)
```

Parameters

native `AndroidJavaObject`

The native object.

allow `bool`

Whether this download may proceed over a metered connection.

Exceptions

`PlatformNotSupportedException`

Thrown if this method is called on a runtime other than Android.

SetAllowedOverRoaming(AndroidJavaObject, bool)

Set whether this download may proceed over a roaming connection. By default, roaming is allowed.

```
public static void SetAllowedOverRoaming(AndroidJavaObject native, bool allow)
```

Parameters

native AndroidJavaObject

The native object.

allow bool

Whether to allow a roaming connection to be used.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

SetDescription(AndroidJavaObject, string)

Set a description of this download, to be displayed in notifications (if enabled).

```
public static void SetDescription(AndroidJavaObject native, string description)
```

Parameters

native AndroidJavaObject

The native object.

description string

The description.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

SetDestinationInExternalFilesDir(AndroidJavaObject, string, string?)

Set the local destination for the downloaded file to a path within the application's external files directory (as returned by `Context.getExternalFilesDir(String)`, aka `Application.persistentDataPath`).

```
public static void SetDestinationInExternalFilesDir(AndroidJavaObject native, string subPath, string? dirType = null)
```

Parameters

native AndroidJavaObject

The native object.

subPath string

The path within the external directory, including the destination filename.

dirType string

The directory type to pass to `Context.getExternalFilesDir(String)`

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

SetDestinationInExternalPublicDir(AndroidJavaObject, string, string)

Set the local destination for the downloaded file to a path within the public external storage directory (as returned by `Environment.getExternalStoragePublicDirectory(String)`).

```
public static void SetDestinationInExternalPublicDir(AndroidJavaObject native,
string subPath, string dirType)
```

Parameters

native AndroidJavaObject

The native object.

subPath string

The path within the external directory, including the destination filename.

dirType string

The directory type to pass to `Environment.getExternalStoragePublicDirectory(String)`

Remarks

For applications targeting Android 10 or above, the `WRITE_EXTERNAL_STORAGE` permission is not needed and the **dirType** must be one of the known public directories like `Environment.DIRECTORY_DOWNLOADS`, `Environment.DIRECTORY_PICTURES`, `Environment.DIRECTORY_MOVIES`, etc.

Exceptions

`PlatformNotSupportedException`

Thrown if this method is called on a runtime other than Android.

SetDestinationUri(AndroidJavaObject, AndroidJavaObject)

Set the local destination for the downloaded file. Must be a file URI to a path on external storage, and the calling application must have the `WRITE_EXTERNAL_STORAGE` permission.

```
public static void SetDestinationUri(AndroidJavaObject native,
AndroidJavaObject uri)
```

Parameters

native `AndroidJavaObject`

The native object.

uri `AndroidJavaObject`

A file Uri indicating the destination for the downloaded file.

Remarks

By default, downloads are saved to a generated filename in the shared download cache and may be deleted by the system at any time to reclaim space.

For applications targeting Android 10 or above, `WRITE_EXTERNAL_STORAGE` permission is not needed and the **uri** must refer to a path within the directories owned by the application (e.g. `Context.getExternalFilesDir(String)`) or a path within the top-level Downloads directory (as returned by `Environment.getExternalStoragePublicDirectory(String)` with `Environment.DIRECTORY_DOWNLOADS`). All non-visible downloads that are not modified in the last 7 days will be deleted during idle runs.

Exceptions

`PlatformNotSupportedException`

Thrown if this method is called on a runtime other than Android.

SetMimeType(AndroidJavaObject, string)

Set the MIME content type of this download. This will override the content type declared in the server's response.

```
public static void SetMimeType(AndroidJavaObject native, string mimeType)
```

Parameters

native `AndroidJavaObject`

The native object.

mimeType `string`

The MIME type.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

SetNotificationVisibility(AndroidJavaObject, NotificationVisibility)

Control whether a system notification is posted by the download manager while this download is running or when it is completed. If enabled, the download manager posts notifications about downloads through the system [NotificationManager](#). By default, a notification is shown only when the download is in progress.

```
public static void SetNotificationVisibility(AndroidJavaObject native,
NotificationVisibility visibility)
```

Parameters

native [AndroidJavaObject](#)

The native object.

visibility [NotificationVisibility](#)

The notification visibility.

Remarks

If set to [Hidden](#), this requires the permission [android.permission.DOWNLOAD_WITHOUT_NOTIFICATION](#).

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

SetRequiresCharging(AndroidJavaObject, bool)

Specify that to run this download, the device needs to be plugged in. This defaults to [false](#)[↗].

```
public static void SetRequiresCharging(AndroidJavaObject native,  
bool requiresCharging)
```

Parameters

native AndroidJavaObject

The native object.

requiresCharging bool

Whether the device is plugged in.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

SetRequiresDeviceIdle(AndroidJavaObject, bool)

Specify that to run, the download needs the device to be in idle mode. This defaults to [false](#)[↗].

```
public static void SetRequiresDeviceIdle(AndroidJavaObject native,  
bool requiresDeviceIdle)
```

Parameters

native AndroidJavaObject

The native object.

requiresDeviceIdle bool

Whether the device need be within an idle maintenance window.

Remarks

Idle mode is a loose definition provided by the system, which means that the device is not in use, and has not been in use for some time.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

SetTitle(AndroidJavaObject, string)

Set the title of this download, to be displayed in notifications (if enabled). If no title is given, a default one will be assigned based on the download filename, once the download starts.

```
public static void SetTitle(AndroidJavaObject native, string title)
```

Parameters

native AndroidJavaObject

The native object.

title string

The title.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.

Enum AndroidInterop.DownloadStatus

Namespace: [Uralstech.UDownloadManager.Native](#)

Possible statuses for a download.

```
[Flags]  
public enum AndroidInterop.DownloadStatus
```

Extension Methods

[AndroidInterop.ToDownloadStatus\(AndroidInterop.DownloadStatus\)](#)

Fields

Failed = 16

The download has failed (and will not be retried).

None = 0

None, invalid status.

Paused = 4

The download is waiting to retry or resume.

Pending = 1

The download is waiting to start.

Running = 2

The download is currently running.

Successful = 8

The download has successfully completed.

Class AndroidInterop.Uri

Namespace: [Uralstech.UDownloadManager.Native](#)

Partial interface for [Uri](#).

```
public static class AndroidInterop.Uri
```

Inheritance

object ← AndroidInterop.Uri

Methods

CreateInstance(string)

Creates a new Uri object.

```
public static AndroidJavaObject CreateInstance(string uri)
```

Parameters

uri string

The HTTP or HTTPS URI.

Returns

AndroidJavaObject

A new instance of Uri.

Remarks

The returned AndroidJavaObject should be disposed when no longer needed.

Exceptions

PlatformNotSupportedException

Thrown if this method is called on a runtime other than Android.